

CANTIDAD DE MOVIMIENTO Y CHOQUES

$$\sum \mathbf{F} = \frac{d\mathbf{p}}{dt}, \quad \mathbf{J} = \int \sum \mathbf{F} d\mathbf{t} = \Delta \mathbf{p}$$

$$\mathbf{r}_{\text{cm}} = \frac{1}{M} \sum m_i \mathbf{r}_i, \quad \mathbf{P} = M \mathbf{v}_{\text{cm}}$$

$$\sum \mathbf{F}_{\text{ext}} = M \mathbf{a}_{\text{cm}}$$

$$\mathbf{P}_{\text{antes}} = \mathbf{P}_{\text{desp}}, \quad K_{\text{antes}} = K_{\text{desp}} \quad (\text{elástico})$$

$$K = \frac{1}{2} M v_{\text{cm}}^2 + K_{\text{rel al CM}}$$

COHETE

$$\mu = -\frac{dM}{dt}, \quad f = \mu |v_r| = I_{\text{sp}} g_0 \mu$$

$$M \mathbf{a} = -\mu \mathbf{v}_r$$

$$\frac{dV}{dt} = \frac{\mu |v_r|}{M_0 - \mu t} - g$$

$$V_f = V_0 + |v_r| \ln \frac{M_0}{M_f} - g t_f$$

$$\Delta V = |v_r| \ln \frac{M_0}{M_f}, \quad \Delta V_{\text{tot}} = \sum_{\text{etapas}} \Delta V_i$$

TRABAJO Y ENERGÍA

$$W = \int \mathbf{F} \cdot d\mathbf{l}, \quad W_{\text{tot}} = \Delta K, \quad P = \mathbf{F} \cdot \mathbf{v}$$

$$\mathbf{F} = -\nabla U, \quad W_{\text{no cons}} = \Delta(K + U)$$

GRAVITACIÓN

$$\mathbf{F}_g = -\frac{GMm}{r^2} \hat{\mathbf{r}}, \quad U = -\frac{GMm}{r}, \quad g = \frac{\mu_T}{R_T^2}$$

$$v_{\text{circ}} = \sqrt{\frac{\mu}{r}}, \quad v_{\text{esc}} = \sqrt{\frac{2\mu}{r}} = \sqrt{2} v_{\text{circ}}$$

MOMENTO ANGULAR (PARTÍCULA)

$$\mathbf{L}_O = \mathbf{r} \times m \mathbf{v}, \quad L = mvr \sin \varphi$$

$$\sum \boldsymbol{\tau}_O = \frac{d\mathbf{L}_O}{dt}$$

$$h = rv_\theta = rv \cos \gamma = \text{cte}, \quad \frac{dA}{dt} = \frac{h}{2}$$

$$r_P v_P = r_A v_A \quad (\mathbf{r} \perp \mathbf{v} \text{ en } P \text{ y } A)$$

DOS CUERPOS

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3} \mathbf{r}, \quad \mu = G(m_1 + m_2)$$

$$m_r = \frac{m_1 m_2}{m_1 + m_2}$$

$$\mathbf{r}_1 = \frac{m_2}{m_1 + m_2} \mathbf{r}, \quad \mathbf{r}_2 = -\frac{m_1}{m_1 + m_2} \mathbf{r}$$

ÓRBITA: CÓNICAS Y KEPLER

$$r = \frac{p}{1 + e \cos \nu}, \quad p = \frac{h^2}{\mu} = a(1 - e^2)$$

$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a}$$

$$v^2 = \mu \left(\frac{2}{r} - \frac{1}{a} \right)$$

$$\frac{1}{2} v_P^2 - \frac{\mu}{r_P} = \frac{1}{2} v_A^2 - \frac{\mu}{r_A}$$

$$r_P = a(1 - e), \quad r_A = a(1 + e)$$

$$a = \frac{r_P + r_A}{2}, \quad e = \frac{r_A - r_P}{r_A + r_P}, \quad b = \sqrt{r_P r_A}$$

$$\tau = \frac{2\pi ab}{h} = 2\pi \sqrt{\frac{a^3}{\mu}}$$

$$v_\perp = \frac{h}{r}, \quad v_r = \frac{\mu}{h} e \sin \nu$$

$$\tan \gamma = \frac{v_r}{v_\perp} = \frac{e \sin \nu}{1 + e \cos \nu}$$

MANIOBRAS: HOHMANN Y FASAJE

$$v_P = \sqrt{\frac{2\mu r_A}{r_P(r_A + r_P)}}, \quad v_A = \frac{r_P}{r_A} v_P$$

$$v_{c1} = \sqrt{\frac{\mu}{r_P}}, \quad v_{c2} = \sqrt{\frac{\mu}{r_A}}$$

$$\Delta v_1 = v_P - v_{c1}, \quad \Delta v_2 = v_{c2} - v_A$$

$$a_t = \frac{r_P + r_A}{2}, \quad t_v = \pi \sqrt{\frac{a_t^3}{\mu}}$$

$$\varphi = 180^\circ - n_2 t_v, \quad n_2 = 360^\circ / \tau_2$$

$$\frac{T'}{T} = 1 - \frac{\Delta \varphi}{360^\circ}, \quad a' = r \left(\frac{T'}{T} \right)^{2/3}$$

HIPÉRBOLA, ESCAPE Y SOI

$$a = \frac{p}{e^2 - 1}, \quad r_p = a(e - 1), \quad \varepsilon = \frac{\mu}{2a}$$

$$\nu_\infty = \arccos\left(-\frac{1}{e}\right), \quad \delta = 2 \arcsin\left(\frac{1}{e}\right)$$

$$v_\infty = \sqrt{\frac{\mu}{a}}, \quad v^2 = v_{\text{esc}}^2 + v_\infty^2, \quad C_3 = v_\infty^2$$

$$r_{\text{SOI}} = R \left(\frac{m_p}{m_s} \right)^{2/5}$$

$$v_\infty = V_{\text{nave}} - V_{\text{planeta}}, \quad v_c = \sqrt{\mu/r_p}$$

$$e = 1 + \frac{r_p v_\infty^2}{\mu}, \quad v_p = \sqrt{v_\infty^2 + \frac{2\mu}{r_p}}$$

$$\Delta v = v_p - v_c = v_c \left(\sqrt{2 + (v_\infty/v_c)^2} - 1 \right)$$

ROTACIÓN ALREDEDOR DE UN EJE FIJO

$$v = r\omega, \quad a_{\text{tan}} = r\alpha, \quad a_{\text{rad}} = \omega^2 r$$

$$I = \sum m_i r_i^2, \quad K = \frac{1}{2} I \omega^2$$

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}, \quad \sum \tau_z = I \alpha_z, \quad L = I \omega$$

$$\sum \boldsymbol{\tau}_{\text{ext}} = \frac{d\mathbf{L}}{dt}, \quad I_1 \omega_1 = I_2 \omega_2$$

$$K = \frac{1}{2} M v_{\text{cm}}^2 + \frac{1}{2} I_{\text{cm}} \omega^2, \quad v_{\text{cm}} = R\omega$$

$$I_P = I_{\text{cm}} + M d^2, \quad \Omega_{\text{prec}} = \frac{M g r}{I \omega}$$

varilla (cm) $ML^2/12$ aro MR^2
 varilla (ext) $ML^2/3$ disco $MR^2/2$
 esfera $2MR^2/5$ esf. hueca $2MR^2/3$

CUERPO RÍGIDO EN 3D

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$$

$$\mathbf{a}_B = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{B/A} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_{B/A})$$

$$\left(\dot{\mathbf{Q}} \right)_{\text{fijo}} = \left(\dot{\mathbf{Q}} \right)_{\text{rot}} + \boldsymbol{\Omega} \times \mathbf{Q}$$

$$\begin{pmatrix} H_x \\ H_y \\ H_z \end{pmatrix} = \begin{pmatrix} I_x & -I_{xy} & -I_{xz} \\ -I_{xy} & I_y & -I_{yz} \\ -I_{xz} & -I_{yz} & I_z \end{pmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix}$$

$$\mathbf{H}_O = \tilde{\mathbf{r}} \times m \tilde{\mathbf{v}} + \mathbf{H}_G$$

$$T = \frac{1}{2} m \tilde{v}^2 + \frac{1}{2} \boldsymbol{\omega} \cdot \mathbf{H}_G$$

$$\sum M_x = I_x \dot{\omega}_x - (I_y - I_z) \omega_y \omega_z \quad (\text{y cicl.})$$

$$\sum M_O = \dot{\varphi} \sin \theta [I' \dot{\psi} + (I' - I) \dot{\varphi} \cos \theta] \hat{\mathbf{f}}$$

$$\theta = 90^\circ : \quad \sum M_O = I' \dot{\varphi} \dot{\psi} \hat{\mathbf{f}}$$

$$\text{libre: } \dot{\varphi} = H/I, \quad \tan \gamma = \frac{I}{I'} \tan \theta$$

VECTOR DE ESTADO Y LAGRANGE

$$\mathbf{r} = \frac{h^2}{\mu} \frac{1}{1 + e \cos \nu} (\cos \nu \hat{\mathbf{p}} + \sin \nu \hat{\mathbf{q}})$$

$$\mathbf{v} = \frac{\mu}{h} [-\sin \nu \hat{\mathbf{p}} + (e + \cos \nu) \hat{\mathbf{q}}]$$

$$\mathbf{h} = \mathbf{r} \times \mathbf{v}, \quad \mathbf{n} = \hat{\mathbf{k}} \times \mathbf{h}$$

$$\mathbf{e} = \frac{1}{\mu} \left[\left(v^2 - \frac{\mu}{r} \right) \mathbf{r} - (\mathbf{r} \cdot \mathbf{v}) \mathbf{v} \right]$$

$$\mathbf{r} = f \mathbf{r}_0 + g \mathbf{v}_0, \quad \mathbf{v} = \dot{f} \mathbf{r}_0 + \dot{g} \mathbf{v}_0$$

$$f = 1 - \frac{\mu r}{h^2} (1 - \cos \Delta \nu), \quad g = \frac{r r_0}{h} \sin \Delta \nu$$

$$\dot{g} = 1 - \frac{\mu r_0}{h^2} (1 - \cos \Delta \nu), \quad f \dot{g} - \dot{f} g = 1$$

TRES CUERPOS RESTRINGIDO

$$\Omega = \sqrt{\mu/r_{12}^3}, \quad \pi_i = \frac{m_i}{m_1 + m_2}$$

$$x_1 = -\pi_2 r_{12}, \quad x_2 = \pi_1 r_{12}$$

$$\ddot{x} - 2\Omega \dot{y} - \Omega^2 x = -\frac{\mu_1}{r_1^3} (x + \pi_2 r_{12}) - \frac{\mu_2}{r_2^3} (x - \pi_1 r_{12})$$

$$\ddot{y} + 2\Omega \dot{x} - \Omega^2 y = -\left(\frac{\mu_1}{r_1^3} + \frac{\mu_2}{r_2^3} \right) y$$

$$L_{4,5} : \left(\frac{r_{12}}{2} - \pi_2 r_{12}, \pm \frac{\sqrt{3}}{2} r_{12} \right)$$

$$r_{\text{Hill}} = r_{12} \left(\frac{m_2}{3m_1} \right)^{1/3}, \quad \frac{m_1}{m_2} + \frac{m_2}{m_1} \geq 25$$

$$C = \frac{v^2}{2} - \frac{\Omega^2 (x^2 + y^2)}{2} - \frac{\mu_1}{r_1} - \frac{\mu_2}{r_2}$$

MARCOS NO INERCIALES

$$\mathbf{v} = \dot{r} \hat{\mathbf{r}} + r \dot{\theta} \hat{\boldsymbol{\theta}}$$

$$\mathbf{a} = (\ddot{r} - r \dot{\theta}^2) \hat{\mathbf{r}} + (r \ddot{\theta} + 2\dot{r} \dot{\theta}) \hat{\boldsymbol{\theta}}$$

$$m \mathbf{a}' = \mathbf{F} - m \mathbf{A}$$

$$m \mathbf{a}_{\text{rel}} = \mathbf{F} - 2m \boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}} - m \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}) - m \dot{\boldsymbol{\Omega}} \times \mathbf{r}$$

CONSTANTES

G	$6,674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$
μ_T	$398\,600 \text{ km}^3/\text{s}^2$
R_T	6378 km
μ_{Luna}	$4903 \text{ km}^3/\text{s}^2$
$d_{\text{T-L}}$	$384\,400 \text{ km}$
μ_{Sol}	$1,327 \times 10^{11} \text{ km}^3/\text{s}^2$
1 UA	$149,6 \times 10^6 \text{ km}$
g_0	$9,80665 \text{ m/s}^2$